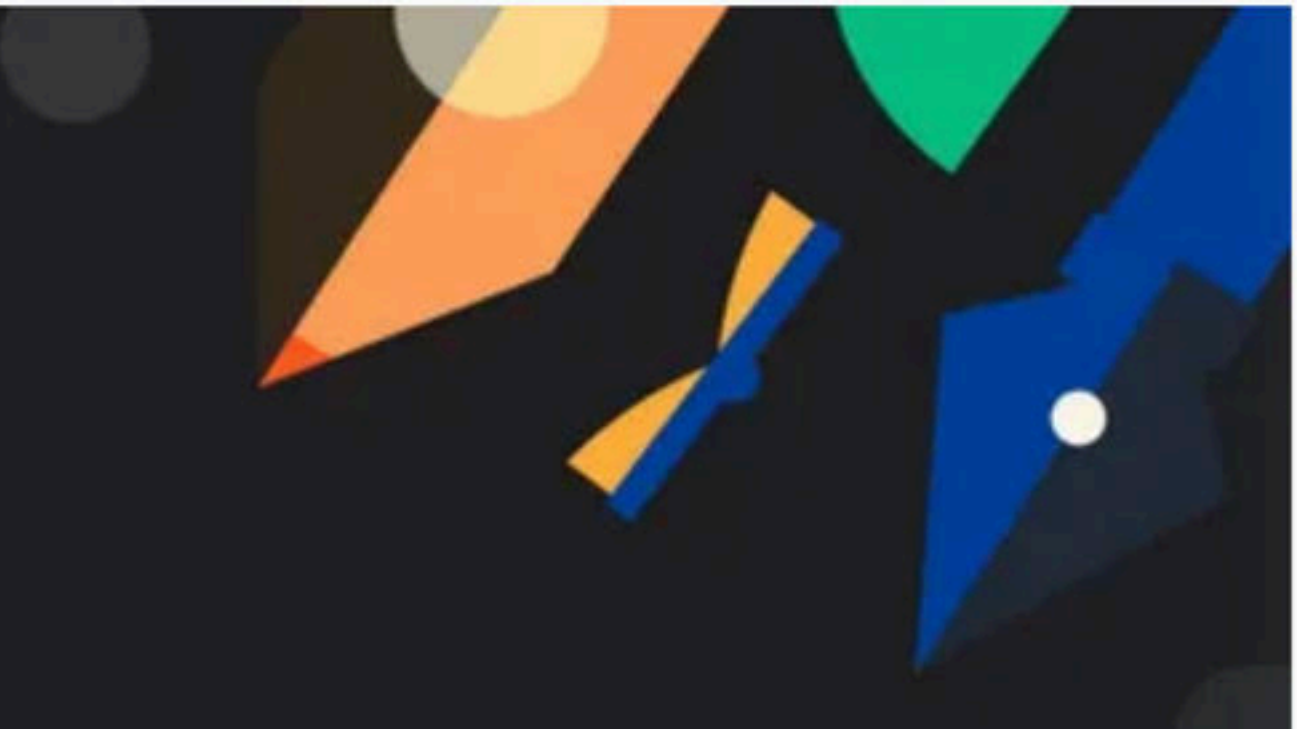




Bit Manipulation BONUS Session

Special class

①

Bits

Operators

0000010
0000100
 <<1

0 → 1
1 → 0

XOR

<<

>>

0 0 → 0
0 1 → 1
1 0 → 0
1 1 → 1

0 0 → 0
0 1 → 1
1 0 → 1
1 1 → 1

0 0 → 0
0 1 → 1
1 0 → 1
1 1 → 0

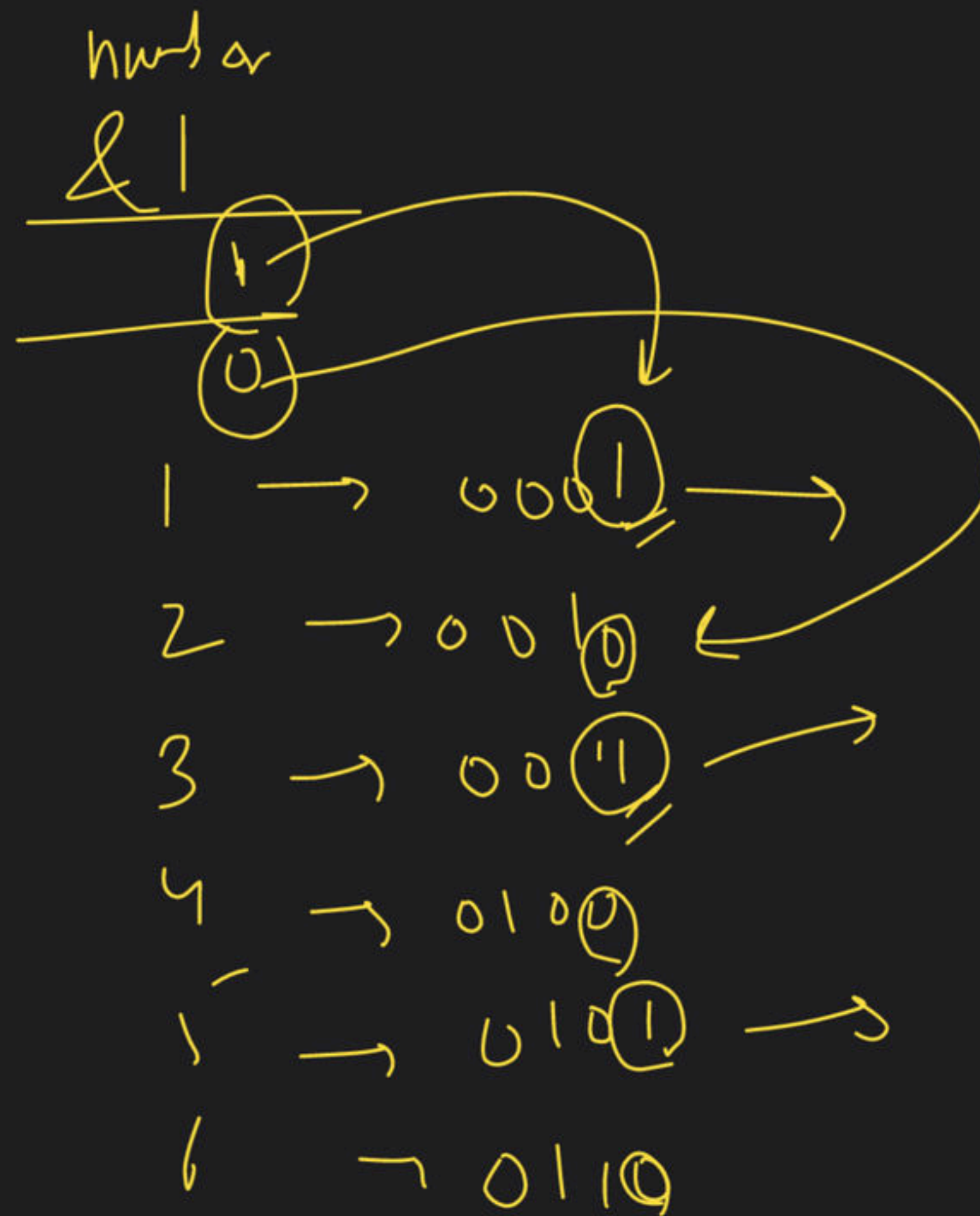
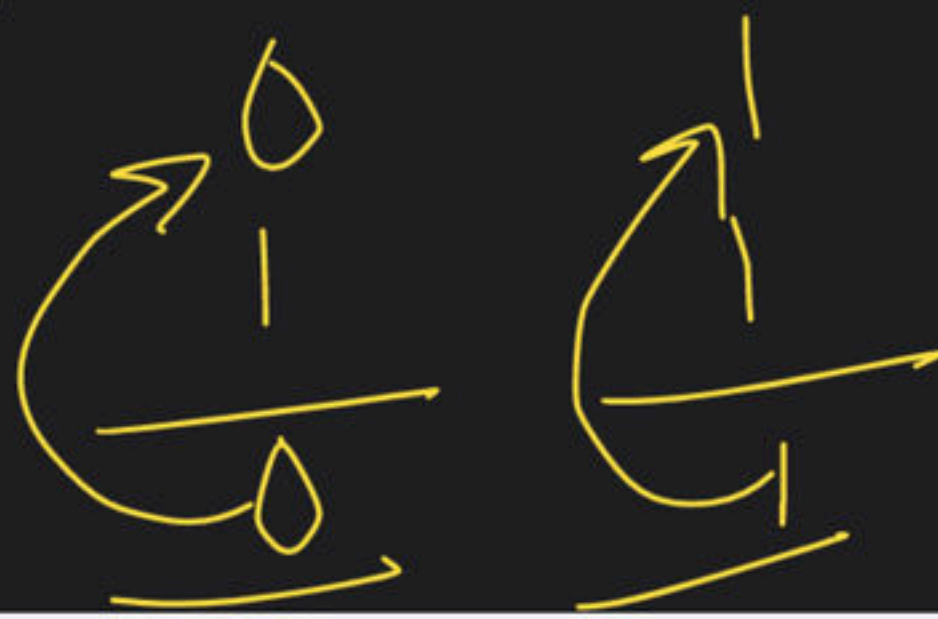
B → D
D → B
Sig-bit

① i/p $\rightarrow$ number $\rightarrow 8$

o/p $\rightarrow$ Even / Odd

check bit $\rightarrow 0 / 1$

&1



g →

0000 — — — 0001000

(21) → mark



→ get i^{th} bit

n
 $\text{mask} < (1 << i)$

$(3) \begin{matrix} 2^3 & 2^2 & 2^1 & 2^0 \\ 4 & 2 & 1 & 0 \end{matrix}$
←

→ $n = 10$

$n \& \text{mask}$

→ $i \rightarrow 3$

→ 0000 - - - 0001

$\ll i$
0000 - - - 001000

0000

00 - -

0000

$\begin{matrix} 3 & 2 & 1 & 0 \\ \downarrow & \downarrow & \downarrow & \downarrow \end{matrix}$

1010

00000000 - - - 001000 → mask

00 - - - 01000 → 1

→ Set n^{th} bit

$$\text{mask} = (1 \ll i)$$

$n \text{ or mask}$

$i = 2$

$n \rightarrow 9$

0000 - - 00 10 01

00000000 100

0000 0000 100

1
↓
0000 - - 000001

$\ll i$

00000000 00

→ clear i^{th} bit

$$\left(1 \ll i \right)$$

$n = 5$

$n = 5$
↓

0000 - - 000 1 0 1

2

1111 - - 110 1 1 1

0000 - - 000 1 0 1

0000 000 - - 0 1000

0000 - - 000 1 0 1

1111 - - 111 0 1 1

0000 - - 000 0 0 1



0000 - - 0001

$\ll 3$

$\checkmark 1 \ll 3$

0000 - - 00001000

5
1111 - - 1110111

→ update ^{ith} bit

n = 10

i = 2

target → 0/1

0000 - 0000 | 0 | 0



set → OK mark

0000 - 0000 | 1 | 0 | 0

0000 - 0000 | 0 | 0 | 0

① clear ith bit

② target < i
OK

→ clear Last i bits

~~1~~
~~00~~
~~0~~

0
~~20~~
0

15 → 0000 00 - 0000 1111
1111 1111 0000
 $i=3$ → 0000 0000 0000 1000

1 → 000000 - - - 0001
-1 → 1111111111111111
1111111111111111 680
 $< i$

→ Check power of 2

$n \& (n-1)$

$n = 18$

2^n

$n \rightarrow$

count set bit

1

1 set Bit

$2 \rightarrow 10$
 $4 \rightarrow 100$
 $8 \rightarrow 1000$
 $16 \rightarrow 10000$
 $32 \rightarrow 100000$

$2^1 \rightarrow 2$

$2^2 \rightarrow 4$

$2^3 \rightarrow 8$

$2^4 \rightarrow 16$

$2^5 \rightarrow 32$

$2^6 \rightarrow 64$

$2^7 \rightarrow 128$

$2^8 \rightarrow 256$

$2^9 \rightarrow 512$

$2^{10} \rightarrow 1024$

n -

$$n = n \& (n-1) \rightarrow \text{count}++$$

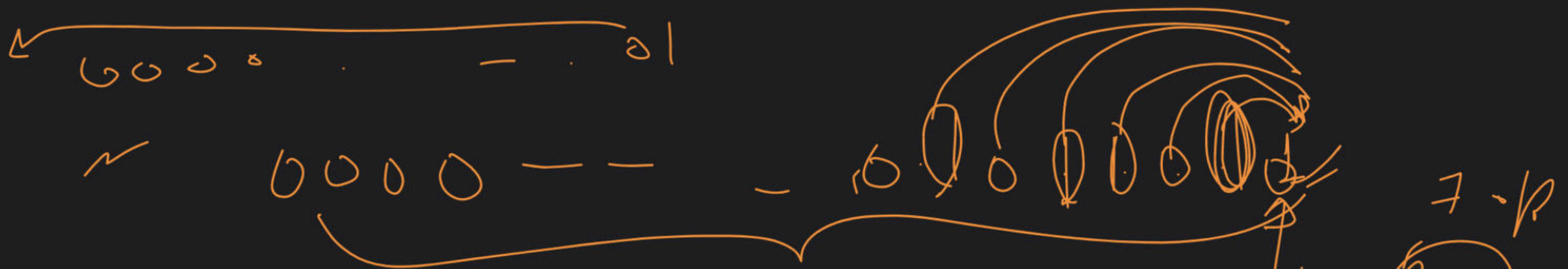
$$n = n \& (n-1) \rightarrow \text{count}++$$

$$n = n \& (n-1) \quad \text{count}++$$

formula $n =$ $n \& (n-1)$

→ Last set bit

remove



$$\begin{cases}
 n = (n \& (n-1)) \longrightarrow \\
 h = (n \& (h-1)) \longrightarrow \\
 h = (n \& (n-1)) \longrightarrow \\
 n = (n \& h-1) \longrightarrow
 \end{cases}
 \rightarrow y_{\text{set}}$$

$n \log(n)$



set bit

2^L

$$n=4$$

0000 - - - 00100

$n-1=3$ & 0000 - - - 00011

00000000

Count set Bit $\rightarrow$

$$h = n \& (n-1)$$

$\rightarrow j^{th} + k$

$$h=0$$

Exponentiation

n/v

pow ()

Bit
Solution

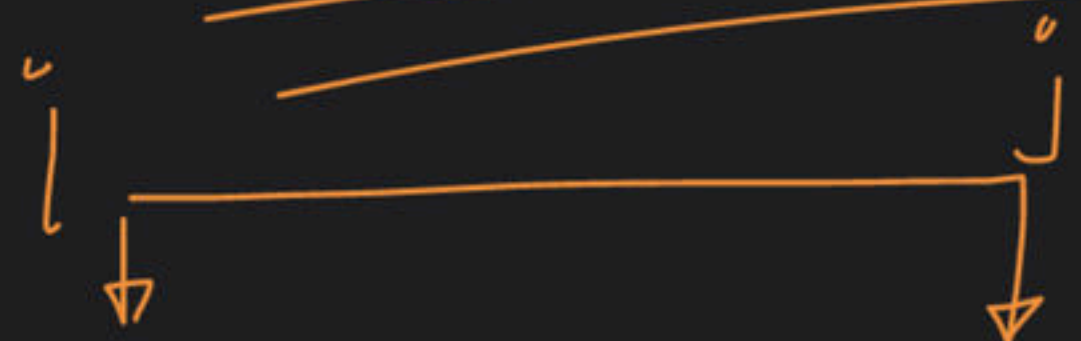
a

for ($i = 0$ — $< s$)
 $ans = ans \times 2$

return

Clear Bits in a Range

1111 0000 0 1111



$a \rightarrow (-1 \ll i)$
 $b \rightarrow \sim (-1 \ll j)$
 $m = a / b$

mask

$-1 \ll i$

1111 0000 0000 1111

0000 0000 0000 1111

$a \rightarrow$ 1111 0000 0000 0000

$b \rightarrow$ 0000 0000 0000 1111

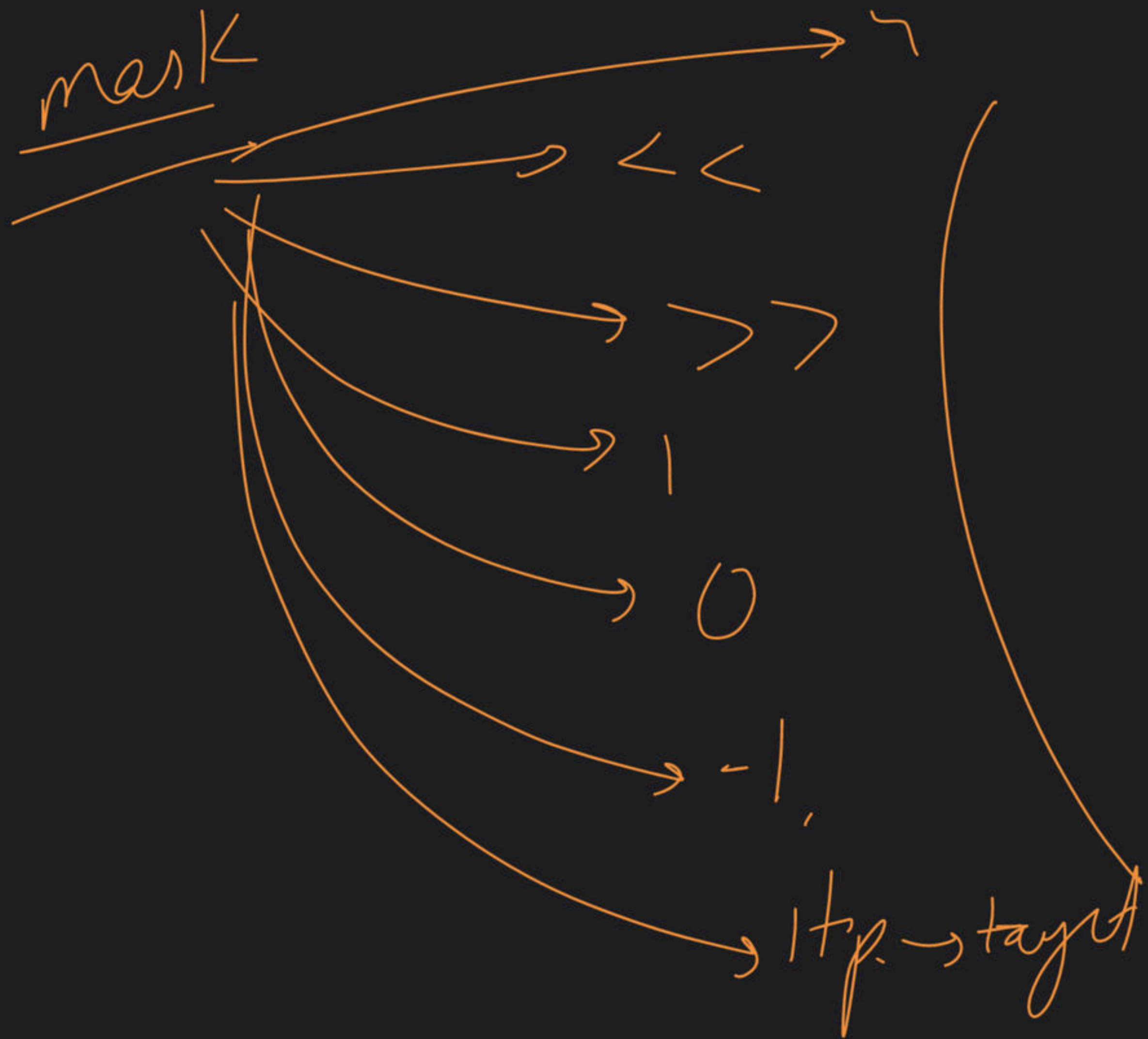
$a \text{ or } b$

1111 0000 0000 1111

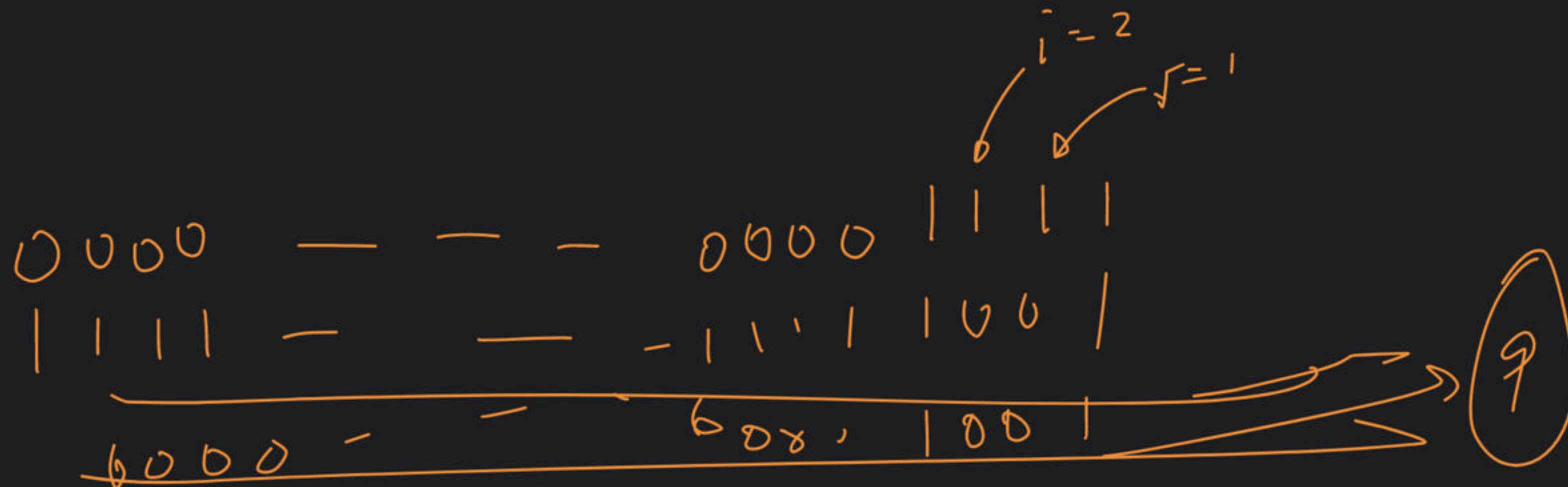
$\hookrightarrow$

00000000	—	—	00111
11111	—	—	111000

$\hookrightarrow (-1 < j)$



n=15



$a \rightarrow (-1 < \textcircled{4}) \rightarrow$ 1111 — — — 1111 1111

$b = \textcircled{4}(-1 < \textcircled{j})$ $a \rightarrow$ 1111 — — — 1111 1000



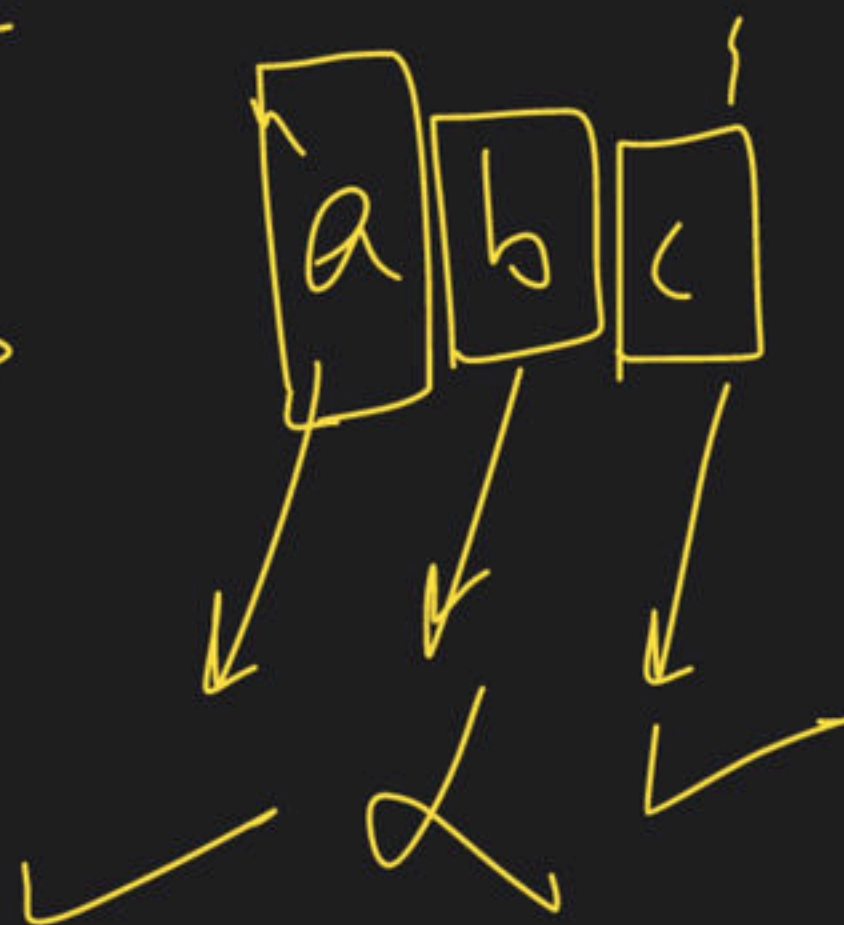
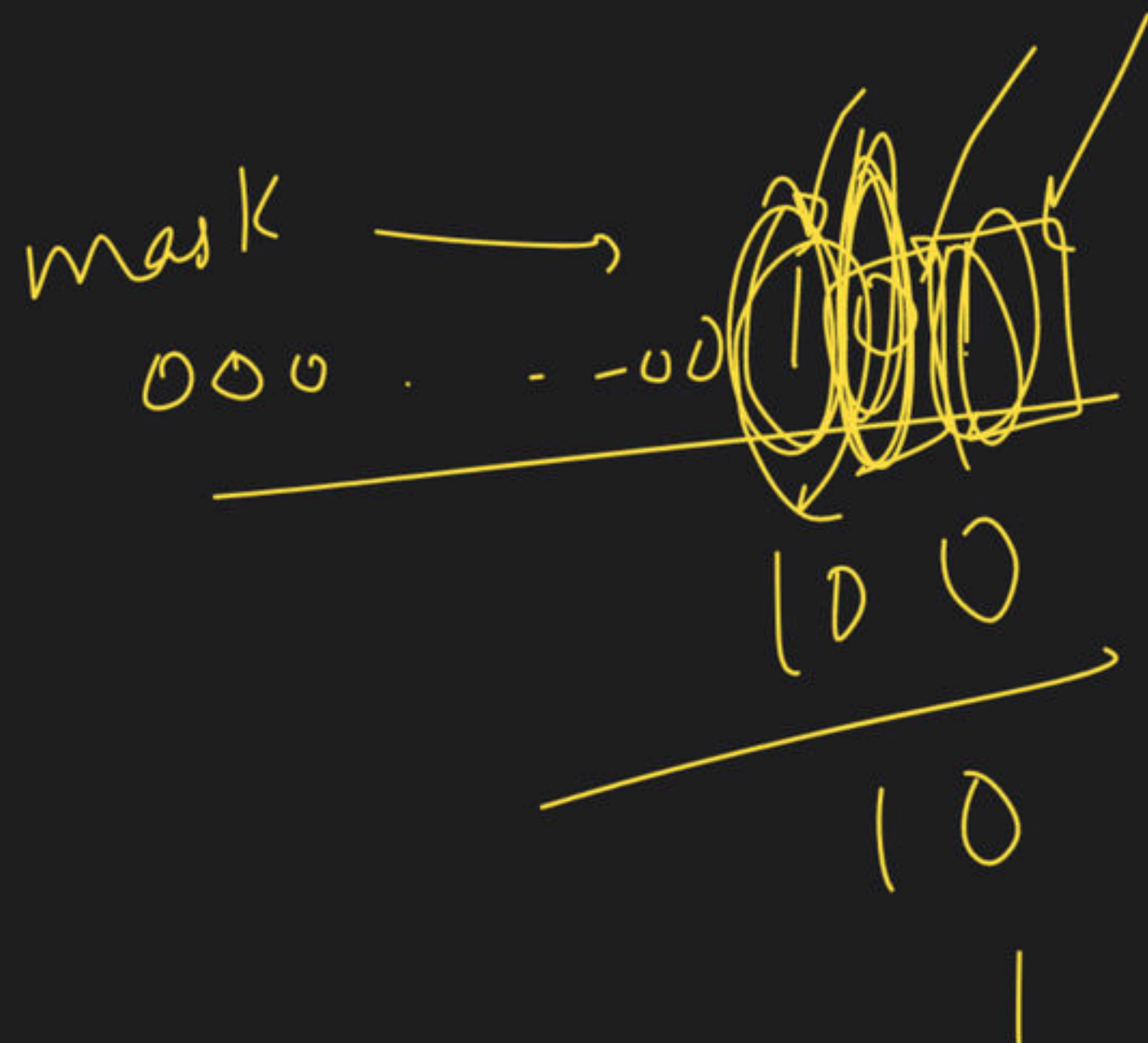
1111 — — — 1111 1111
1111 — — — 1111 1110 $\rightarrow$

→ Subsequence
 str → "abc"

$3 \rightarrow 2^3 \rightarrow 8$

	a	b	c	
0	0	0	0	" "
1	0	0	1	"c"
2	0	1	0	"b"
3	0	1	1	"bc"
4	1	0	0	"a"
5	1	0	1	"ac"
6	1	1	0	"ab"
7	1	1	1	"abc"

mask $\leftrightarrow 0 \rightarrow 7$
 $0 \rightarrow 2^3 - 1$
 $0 \rightarrow < 2^3$



check

✓















